Numerical Methods

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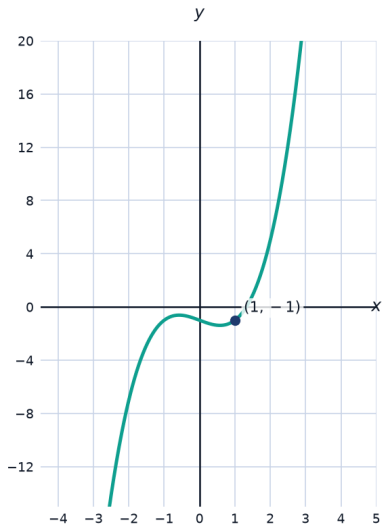
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = x^3 - 4x - 2$ has a root between $x = 2$ and $x = 3$.

2 [Foundational] • Target: Interval bisection

Using $f(x) = x^3 - x - 1$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval containing the root.



3 [Foundational] • Target: Interval bisection

Continuing from the previous problem, evaluate $f(1.25)$ and give a narrower interval for the root.

4 [Foundational] • Target: Interval bisection

Using $f(x) = \ln x - 2 + x$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval for the root.

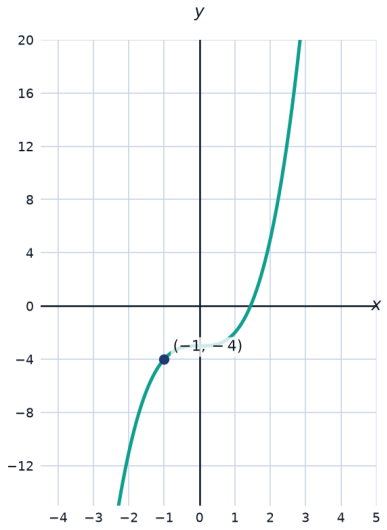
5 [Foundational] • Target: Fixed-point iteration

The equation $x^3 - 4x - 2 = 0$ rearranges to $x = \sqrt[3]{4x + 2}$. Using $x_0 = 2$, find x_1, x_2, and x_3, to four decimal places.

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6 [Foundational] • Target: Convergence and divergence of iterations

Explain why the iteration $x_{n+1} = x_n^3 - 3$ (i.e. $g(x) = x^3 - 3$), starting from $x_0 = 1.5$, would fail to converge to a root of $x^3 - x - 3 = 0$ near $x = 1.5$.



7 [Foundational] • Target: Convergence and divergence of iterations

Using $g(x) = 5 - \frac{2}{x}$ from the earlier iteration, find $g'(x)$ and evaluate $|g'(x)|$ at $x = 4.56$ (near the root), to explain why the iteration converges there.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^2 - 3x - 1 = 0$ into the form $x = g(x)$ by making the linear term the subject.

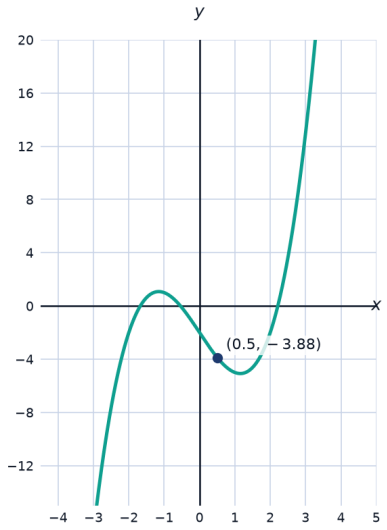
9 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^3 + x - 1 = 0$ into the form $x = \frac{1}{x^2 + 1}$, and verify algebraically that this is equivalent to the original equation.

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10 [Intermediate] • Target: The Newton-Raphson method

Use one iteration of the Newton-Raphson method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 4x - 2$, to four decimal places.



11 [Intermediate] • Target: The Newton-Raphson method

Continuing from $x_1 = 2.25$ in the previous problem, find x_2, to four decimal places.

12 [Intermediate] • Target: The Newton-Raphson method

Use the Newton-Raphson method with $x_0 = 2.5$ to find x_1 for $f(x) = x^2 - 7$ (approximating $\sqrt{7}$), to four decimal places.

13 [Intermediate] • Target: The Newton-Raphson method

Use the Newton-Raphson method with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 2x - 5$, to four decimal places.

14 [Intermediate] • Target: Further Newton-Raphson: rapid convergence and failure cases

Using $x_1 = 2.65$ from the previous problem for $f(x) = x^2 - 7$, perform a second iteration to find x_2, to four decimal places, and compare with $\sqrt{7} \approx 2.6458$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

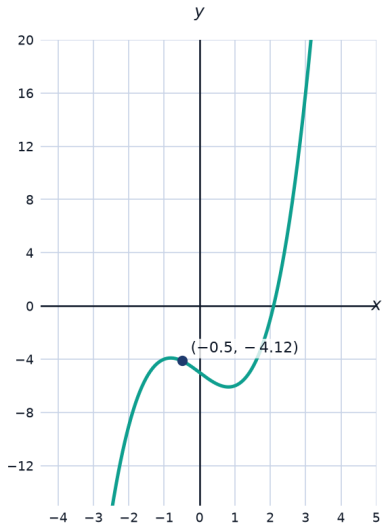
15 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 11.01 Differential Equations

Explain why $x_0 = 1$ is an unsuitable starting point for the Newton-Raphson method applied to $f(x) = x^3 - 3x + 1$.

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16 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 12.01 Poisson Distribution

Continuing from $x_1 = 2.1$ for $f(x) = x^3 - 2x - 5$, perform a second Newton-Raphson iteration to find x_2, to four decimal places.



17 [Challenging] • Target: Error bounds from bisection • Cross-Lesson: 13.01 Continuous Random Variables

A root of $f(x) = x^3 - 4x - 2$ lies in $(2.21, 2.22)$, since $f(2.21) \approx -0.046$ and $f(2.22) \approx 0.061$. State the midpoint estimate and the maximum possible error.

18 [Challenging] • Target: Error bounds from bisection • Cross-Lesson: 14.01 Sampling & Estimation

Given that the root of $f(x) = x^3 - 4x - 2$ is trapped in the interval $(2.2141, 2.2144)$ after further bisection, state the root correct to three decimal places, justifying your answer.

19 [Challenging] • Target: A well-designed word problem: designing a support strut • Cross-Lesson: 15.01 Hypothesis Testing

This question has four parts. An engineer models the cross-sectional profile of a support strut by $f(x) = x^3 - 6x^2 + 9x - 2$ (metres) and needs to find where it crosses the x-axis. (a) Show by a change of sign that a root lies between $x = 0$ and $x = 1$. (b) Use the Newton-Raphson method with $x_0 = 0.5$ to find x_1, to four decimal places. (c) Perform a second iteration to find x_2, to four decimal places. (d) Perform a third iteration to find x_3, and state the root to three decimal places.

20 [Challenging] • Target: Interval bisection • Cross-Lesson: 2.01 Exponentials & Logarithms

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Locating roots by a change of sign

Using the Newton-Raphson method with $f(x) = x^2 - 18$ and $x_0 = 6$, find x_1.

a. $-\frac{9}{2}$

b. $\frac{9}{2}$

c. 6

d. $\frac{11}{2}$

22 [Foundational] • Target: Interval bisection

The unique real root α of the equation $x^3 + 2x - k = 0$, where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$

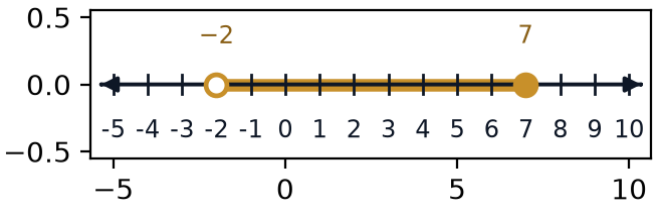
b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

23 [Foundational] • Target: Fixed-point iteration

An interval of width 32 containing a root must be reduced to width at most 1 using repeated bisection. Find the minimum number of bisections required.



a. 7

b. 4

c. 6

d. 5

24 [Foundational] • Target: Convergence and divergence of iterations

An interval of width 8 containing a root is bisected 3 times. Find the width of the resulting interval.

a. 2

b. $\frac{13}{12}$

c. $\frac{8}{3}$

d. 1

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25 [Foundational] • Target: Rearranging equations for iteration

A curve passes through the point $\left(1, \frac{1}{2}\right)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y^2(1 - \ln x)}{x^2}$ for $x > 0$. Determine the exact coordinates of the stationary point of the curve and its nature.

a. $\left(e, \frac{e}{2e-1}\right)$, which is a local maximum

b. $\left(e, \frac{1}{2e-1}\right)$, which is a local maximum

c. $\left(e, \frac{e}{2e+1}\right)$, which is a local minimum

d. $\left(\frac{1}{e}, \frac{1}{2e+1}\right)$, which is a local minimum

26 [Foundational] • Target: The Newton-Raphson method

The continuous function f is defined for all $x \in \mathbb{R}$ by $f(x) = 2^x + 3x - 11$. Between which pair of consecutive integers does a root of the equation $f(x) = 0$ lie?

a. 2 and 3

b. 1 and 2

c. 3 and 4

d. 0 and 1

27 [Foundational] • Target: Further Newton-Raphson: rapid convergence and failure cases

A confidence interval for the mean uses $z = 1$, $\sigma = 4$, $n = 9$. Find the margin of error $E = z\sigma/\sqrt{n}$.

a. $\frac{7}{3}$

b. $\frac{4}{3}$

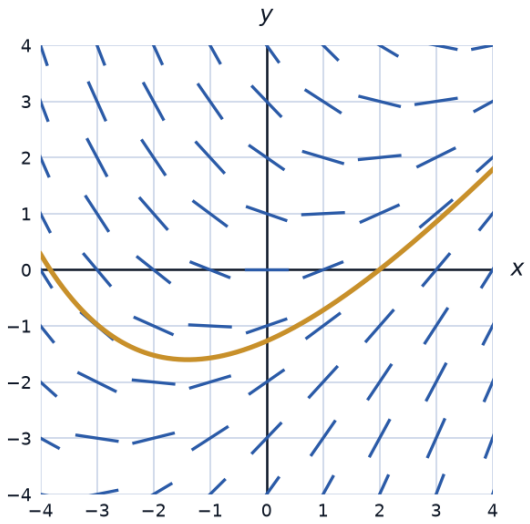
c. $\frac{17}{12}$

d. $\frac{8}{3}$

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28 [Foundational] • Target: Error bounds from bisection

Which differential equation correctly models the statement: "The rate of increase of the number of infected people N is proportional to the product of N and the number of remaining susceptible people $(S - N)$, for fixed population S."?



a. $\frac{dm}{dt} = -km$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dT}{dt} = -k(T - T_e)$

d. $\frac{dN}{dt} = kN(S - N)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: designing a support strut

A sequence is defined by $x_{n+1} = \frac{2x+6}{2}$ with $x_0 = -2$. Find x_1.

a. 2

b. -2

c. 1

d. -1

30 [Intermediate] • Target: Locating roots by a change of sign

A convergent sequence satisfies $x_{n+1} = \frac{2x+2}{4}$. Find its limit L.

a. 1

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. 2

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31 [Intermediate] • Target: Interval bisection

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 25$ and $T_0 = 41$. Find T at the time t when $e^{-kt} = 1/4$.

a. 41

b. 21

c. 29

d. 30

32 [Intermediate] • Target: Fixed-point iteration

The function f is defined on the open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by $f(x) = \frac{2 - \sin x}{\cos x} + \frac{\cos x}{2 - \sin x}$. Determine the exact minimum value of $f(x)$.

a. $\frac{4\sqrt{3}}{3}$

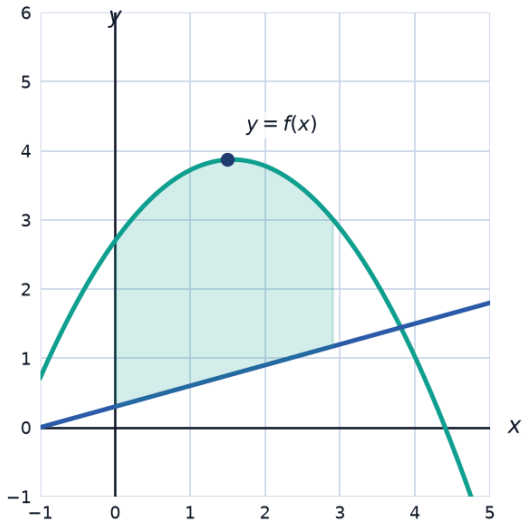
b. 2

c. $\frac{5}{2}$

d. $\frac{3\sqrt{3}}{2}$

33 [Intermediate] • Target: Convergence and divergence of iterations

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.



a. $\frac{4\sqrt{e} - 2e - 1}{2e}$

b. $\frac{4\sqrt{e} - 2e + 1}{2e}$

c. $\frac{2e - 3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e} - 2e - 1}{2e}$

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34 [Intermediate] • Target: Rearranging equations for iteration

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

35 [Intermediate] • Target: The Newton-Raphson method

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. 44

b. 43

c. 11

d. 22

36 [Intermediate] • Target: Further Newton-Raphson: rapid convergence and failure cases

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k - y}{2x + 1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1

b. 11

c. $\frac{5}{4}$

d. 5

37 [Intermediate] • Target: Error bounds from bisection

A curve has equation $y = x^2 e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$

b. $6e^3$

c. $3e^3$

d. $2e^3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: designing a support strut • Cross-Lesson: 14.01 Sampling & Estimation

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

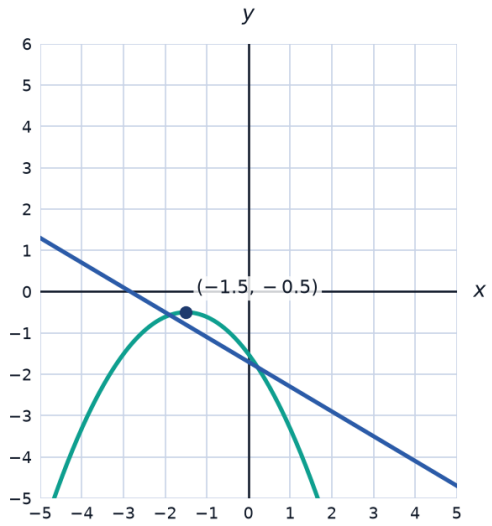
b. $\frac{1}{13}(-3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k})$

c. $\frac{1}{169}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

d. $\frac{1}{19}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

39 [Challenging] • Target: Locating roots by a change of sign • Cross-Lesson: 15.01 Hypothesis Testing

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.



a. 7

b. $\sqrt{31}$

c. $\sqrt{69}$

d. 49

40 [Challenging] • Target: Interval bisection • Cross-Lesson: 2.01 Exponentials & Logarithms

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

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41 [Challenging] • Target: Fixed-point iteration • Cross-Lesson: 3.01 Advanced Trigonometry

The continuous random variable X has probability density function $f(x)$ defined by \[f(x) = \begin{cases} kx & 0 \le x 1 \mid X < 2.5) \).

a. $\frac{9}{11}$

b. $\frac{10}{11}$

c. $\frac{3}{4}$

d. $\frac{9}{10}$

42 [Challenging] • Target: Convergence and divergence of iterations • Cross-Lesson: 4.01 Differentiation

A curve satisfies $x^2 + y^2 = 25$. Find $\frac{dy}{dx}$ at the point $(3, -4)$.

a. $-\frac{3}{4}$

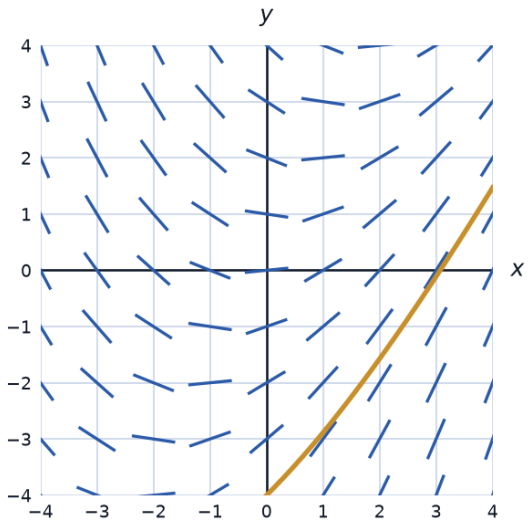
b. $\frac{3}{4}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

43 [Challenging] • Target: Rearranging equations for iteration • Cross-Lesson: 5.01 Integration

Which differential equation correctly models the statement: "The rate of decrease of the temperature T of an object is proportional to the difference between T and the surrounding temperature T_e."?



a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$

c. $\frac{dN}{dt} = kN(S - N)$

d. $\frac{dm}{dt} = -km$

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44 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 6.01 Sequences & Series

Which differential equation correctly models the statement: "The rate of increase of the volume V of a balloon is proportional to the square of its radius r."?

a. $\frac{dT}{dt} = -k(T - T_e)$

b. $\frac{dm}{dt} = -km$

c. $\frac{dV}{dt} = kr^2$

d. $\frac{dP}{dt} = kP$

45 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 7.01 Vectors

Which differential equation correctly models the statement: "The rate of decrease of a radioactive mass m is proportional to m."?

a. $\frac{dN}{dt} = kN(S - N)$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dP}{dt} = kP$

d. $\frac{dm}{dt} = -km$

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